KERALA POLYTECHNIC DIPLOMA - SEMESTER 2 LAB

Fundamentals of Electrical and Electronics Engineering Lab

A full practical handbook for Course Code **2039**. This file is written like a lab manual plus revision book: safety, instruments, procedures, observation tables, circuit logic, formulas, troubleshooting, viva questions and lab-test preparation.

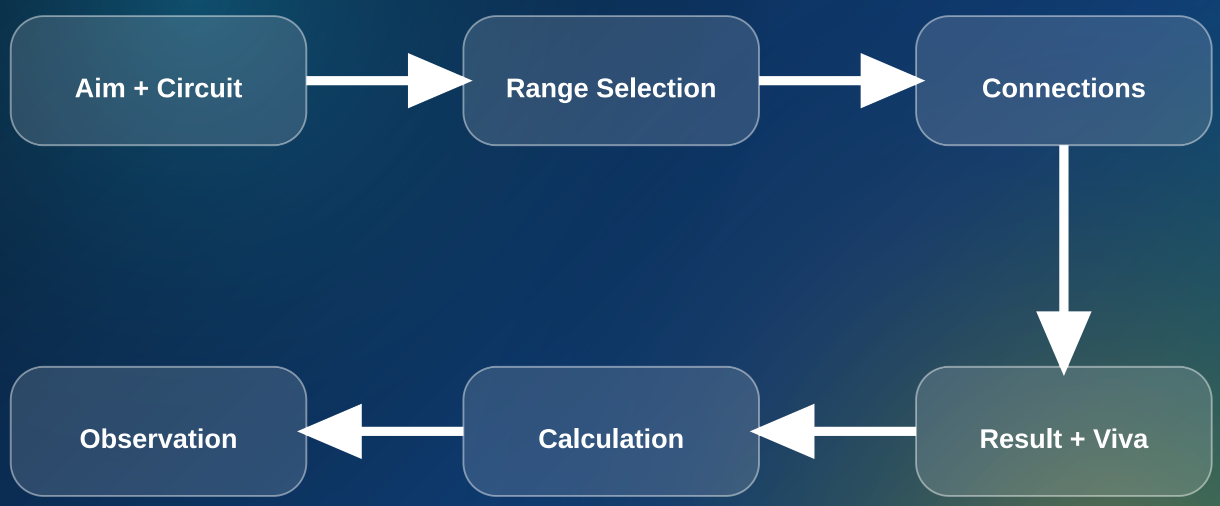
COURSE CODE
2039

CREDITS
No Credit

PERIODS
45 Practical Hours

CATEGORY
Engineering Science

Lab workflow



Hard rule: never energize a circuit until the circuit, range, polarity and fuse rating are verified.

OFFICIAL SYLLABUS CONVERTED TO HANDBOOK

Syllabus structure and practical plan

The course objective is to acquire measurement and testing skills using multimeter, DC voltmeter, ammeter, wattmeter, energy meter and galvanometer, and to experiment on active/passive components and logic gates.

CO	Outcome	Hours	Main practical area
CO1	Make use of various meters to measure basic parameters of an electric circuit.	9	Simple circuit, voltage/current measurement, series and parallel verification
CO2	Identify methods to measure power and energy in electric circuit.	9	Wattmeter, energy meter, lead acid battery maintenance
CO3	Identify components and electronic equipment used for experiments.	9	Passive/active components, instruments, CRO measurement, component testing
CO4	Experiment with rectifier circuits, regulated supply and logic gates.	12	HWR, center-tapped, bridge rectifier, regulator IC, truth tables
Lab tests	Internal practical evaluation	6	Lab Test I and Lab Test II

□ □□□□□□□ theory □□□□□□□ □□□□□□□□□ □□□□. Instrument range, polarity, connection neatness, observation table, calculation and result writing □□□□□□□ □□□□□□□□ important □□□.

Outcomes, PO mapping and assessment thinking

This is a no-credit laboratory course, but the skill value is high. Weak students usually fail because of careless wiring, wrong range selection, incomplete observation tables and poor viva answers.

CO-PO mapping

CO1 is strongly linked to PO1 and PO4. CO2, CO3 and CO4 are strongly linked to PO1. In plain language: this lab mainly builds basic engineering knowledge and practical problem solving.

✓ PO1: Engineering knowledge through circuits and devices.

✓ PO4: Investigation skill through observation, measurement and verification.

✓ Lab test focus: correct circuit, correct instrument, correct result.

How marks are usually lost

- Connecting ammeter in parallel or voltmeter in series.
- Not starting from the highest meter range.

- Ignoring multiplication factor in analog meters.
- Drawing circuit without fuse/switch/source/load labels.
- Not writing units: V, A, W, Wh, Hz, ms, ohm.
- Not switching off before changing circuit wiring.

READ BEFORE DOING ANY EXPERIMENT

Electrical and electronics lab safety

General safety rules

1. Understand the circuit and identify supply, switch, fuse, load and measuring points.
2. Keep the supply OFF while wiring or changing connections.
3. Use correct voltage/current rating of instruments, wires, lamps, diodes, capacitors and regulator IC.
4. Check polarity for DC meters, electrolytic capacitor, LED, diode and regulated power supply.
5. Do not touch exposed conductors. Use insulated leads and avoid loose strands.
6. Inform the instructor immediately if there is smell, smoke, overheating, sparking or abnormal sound.

Brutal warning

Do not treat low-voltage lab as harmless. A wrong connection can destroy meters, short the supply, damage rectifier diodes or heat a capacitor. Your first job is not finishing fast; it is making a safe, verifiable circuit.

Tools, instruments and components

Electrical meters

- Multimeter - voltage, current, resistance, diode/continuity.
- DC voltmeter - connected in parallel.
- DC ammeter - connected in series.
- Wattmeter - pressure coil parallel, current coil series.

- Energy meter - energy in Wh/kWh.
- Galvanometer - sensitive current indication.

Electronic equipment

- Regulated DC power supply.
- Function generator.
- CRO/DSO for waveform measurement.
- Breadboard.
- Transformer and rectifier trainer board.
- Logic gate IC trainer or breadboard setup.

Components

- Resistors, capacitors, inductors and transformer.
- LEDs and diodes.
- Filter capacitor.
- Regulator IC: 7805/7809 or equivalent.
- Logic ICs: AND, OR, NOT, NAND, NOR, XOR.
- Lamps with different power ratings.

Standard lab record format

Use this format for every experiment. A beautiful record with wrong circuit is useless. A correct circuit with incomplete observation is also not acceptable.

Record sequence

1. Aim
2. Apparatus and components
3. Theory in 5 to 8 lines
4. Circuit diagram / connection diagram
5. Procedure
6. Observation table

7. Calculation
8. Result and conclusion
9. Precautions
10. Viva questions

Observation rule

Never write only one trial unless the experiment specifically asks one reading. Minimum good practice is 3 readings or clear before/after comparison.

> Observation table blank □□□□□□□□. Reading + unit + calculation step □□□□□. Viva □□□□□□□ □□□□□□□ □□□□□□□ □□□□□ explain □□□□□□□ □□□□□□.

CO1 - 9 HOURS

Basic electric circuits and meter measurement

Experiments M1.01 to M1.03 train the basic skills every electrical/electronics student must master: circuit assembly, range selection, meter reading, multiplication factor and series/parallel behavior.

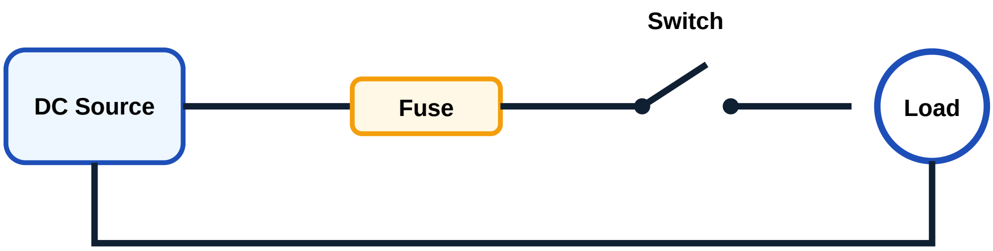
M1.01 Assemble and test a simple electric circuit

Aim: Assemble a circuit containing source, switch, fuse and load, then verify the operation.

Procedure

1. Draw the circuit with source, fuse, switch and lamp/load.
2. Check fuse continuity and load rating.
3. Connect the circuit with supply OFF.
4. Ask for verification before energizing.
5. Switch ON and observe whether the load operates normally.
6. Switch OFF and record result.

Key learning: A fuse is not decoration. It is the protection element placed in series with the live/supply line.



M1.02 Measure voltage and current in a DC circuit

Aim: Select proper meter, range and connection method to measure DC voltage and current.

Meter	Connection	Resistance	Wrong connection effect
Voltmeter	Parallel	Very high	If connected in series, circuit may not work.
Ammeter	Series	Very low	If connected in parallel, meter fuse may blow.
Multimeter	Depends on mode	Mode dependent	Wrong jack/range can damage meter.

Analog meter multiplication factor

$$\text{Actual reading} = \text{Scale reading} \times \text{Range factor}$$

If the scale is 0-10 and selected range is 0-50 V, a pointer at 6 means 6 x 5 = 30 V.

M1.03 Verify voltage and current division

Use three lamps of different power ratings in series and parallel circuits. Compare brightness, voltage and current.

Series circuit

- Same current flows through all loads.
- Voltage divides across loads.
- Total resistance increases.
- If one lamp opens, all go OFF.

Parallel circuit

- Same voltage across all loads.
- Current divides through branches.
- Total resistance decreases.
- If one lamp opens, others can remain ON.

Series: $R_T = R1 + R2 + R3$

Parallel: $1/R_T = 1/R1 + 1/R2 + 1/R3$

CO1 observation table templates

Experiment	Trial	Voltage range	Voltage reading	Current range	Current reading	Remarks
DC circuit	1					
DC circuit	2					
Series lamps	1					
Parallel lamps	1					

CO2 - 9 HOURS + LAB TEST I

Power, energy and battery maintenance

Power tells the rate of energy use. Energy meter tells accumulated electrical energy. Battery maintenance gives practical industrial relevance.

M2.01 Measure power using wattmeter

A wattmeter has two coils: current coil and pressure coil. Current coil is connected in series with the load; pressure coil is connected across the load.

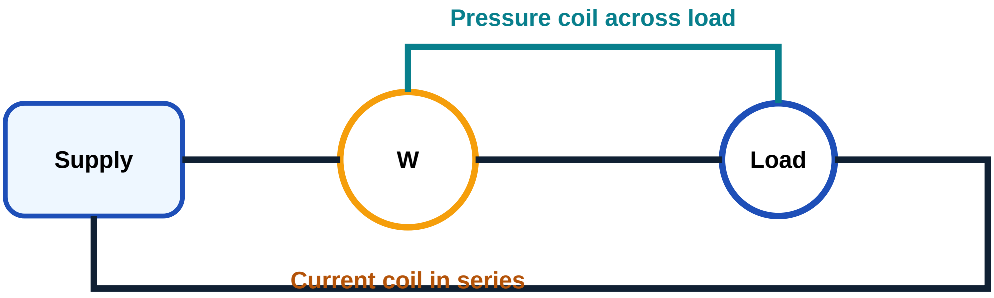
1. Identify M, L, C, V or equivalent wattmeter terminals from the meter.
2. Connect current coil in series with the load.
3. Connect pressure coil across the load.

4. Choose suitable range and switch ON only after verification.

5. Read wattmeter and apply multiplication factor if applicable.

Power, $P = V I \cos\phi$

For DC or pure resistive load, cos phi = 1, so P = V I.



Wattmeter connection concept.

M2.02 Measure energy using energy meter

Energy meter measures electrical energy consumed by a circuit. In lab, energy can be calculated from meter constant, number of revolutions/pulses and time.

Energy = Power x Time

Energy in kWh = (Watt x Hour) / 1000

If meter constant = K rev/kWh, energy for N revolutions = N / K kWh

Common viva: What is one unit of electrical energy? Answer: 1 kWh = 1000 watt-hour.

M2.03 Lead acid battery maintenance

- ✓ Check electrolyte level; plates must remain covered.
- ✓ Use only distilled water for topping up.
- ✓ Clean terminals and apply petroleum jelly if required.
- ✓ Check specific gravity using hydrometer where available.

- ✓ Do not short battery terminals.
- ✓ Avoid sparks near charging battery due to hydrogen gas formation.

> Battery maintenance □□□□□□□ □□□□□ cleaning □□□□□. Electrolyte level, terminal condition, charging safety, ventilation □□□□□□□ □□□□□□□□□□□□□.

CO2 observation table templates

Load	Voltage V	Current I	Wattmeter reading W	Power by VI	Difference	Remarks
Lamp load						
Resistive load						

Meter constant	No. of revolutions/pulses	Time	Calculated energy	Power	Remarks

CO3 - 9 HOURS

Component identification and electronic instruments

This module develops the practical ability to identify components, use breadboards and measure waveforms using CRO.

M3.01 Identify passive components and breadboard

Passive components do not provide power gain. In this lab you identify resistors, capacitors, inductors, transformers and LED, and use a breadboard correctly.

Component	Identification points	Testing method
Resistor	Color code / printed value	Ohmmeter
Capacitor	Capacitance, voltage rating, polarity	Capacitance meter / charging test
Inductor	Coil body, rating	Continuity and inductance meter
Transformer	Primary/secondary voltage rating	Continuity and output voltage test
LED	Anode/cathode, color	Diode mode or series resistor test

Resistor color code quick method

Remember digits: Black 0, Brown 1, Red 2, Orange 3, Yellow 4, Green 5, Blue 6, Violet 7, Grey 8, White 9.

> Resistance = First two digits x $10^{\text{multiplier}}$ ohm

Example: Brown-Black-Red = 10 x 10^2 = 1000 ohm = 1 k ohm.

> Do not confuse multiplier band with tolerance band. Gold/silver usually appears near the end and indicates tolerance.

M3.02 Demonstrate instruments

- **Analog ammeter/voltmeter:** pointer based reading, parallax error possible.
- **Digital multimeter:** direct numerical reading, range and jack selection important.
- **Function generator:** produces sine, square, triangular and pulse waveforms.
- **DC power supply:** regulated adjustable output for circuits.
- **CRO:** waveform display with voltage/time measurement.

M3.03 CRO measurement of sine wave

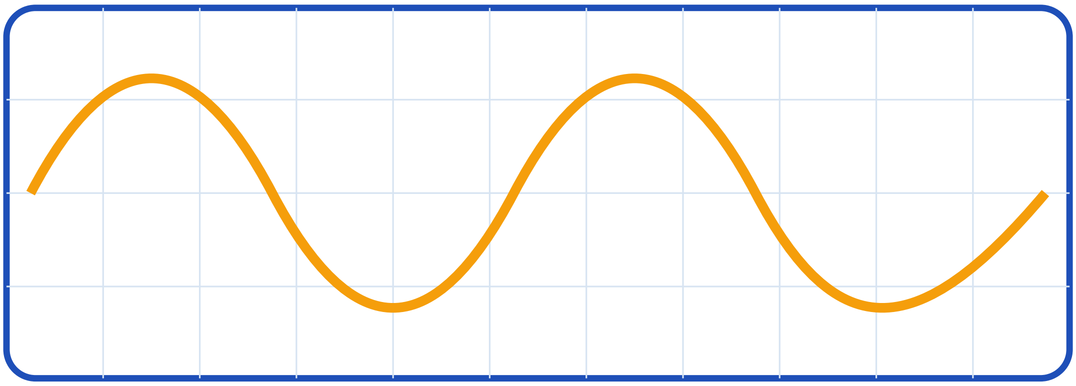
Use CRO vertical divisions to measure amplitude and horizontal divisions to measure time period.

> Peak voltage, Vp = Vertical divisions x Volts/div

> Peak-to-peak voltage, Vpp = Total vertical divisions x Volts/div

> Time period, T = Horizontal divisions for one cycle x Time/div

> Frequency, f = 1/T



One cycle = horizontal divisions x time/div

CRO waveform measurement idea.

M3.04 Test active and passive components

- ✓ Resistor: compare measured resistance with rated value and tolerance.
- ✓ Diode: forward resistance low, reverse resistance high.
- ✓ LED: glows in forward bias with series resistor.
- ✓ Capacitor: observe charge/discharge behavior; electrolytic polarity matters.
- ✓ Transistor: test junctions like two diodes using multimeter diode mode.

CO4 - 12 HOURS + LAB TEST II

Rectifiers, regulated power supply and logic gates

This section connects electronics theory with practical waveform observation and digital logic verification.

M4.01 Half wave rectifier with and without filter

Half wave rectifier uses one diode and converts only one half cycle of AC into pulsating DC. With capacitor filter, ripple reduces and average DC increases.

1. Connect transformer secondary to diode and load.
2. Observe input and output waveform on CRO.
3. Add capacitor filter across load with correct polarity.

4. Measure ripple and calculate ripple factor.

Ripple factor, $r = V_{r(ac\ rms)} / V_{dc}$

Ideal HWR ripple factor without filter approximately 1.21

M4.02 Center tapped full wave rectifier

It uses a center tapped transformer and two diodes. Both half cycles are used, so ripple frequency is twice the supply frequency.

Ripple frequency for full wave rectifier = 2f

Ideal full wave ripple factor without filter approximately 0.482

Important: diode orientation must match the center tapped circuit. Wrong diode direction gives no proper rectification.

M4.03 Bridge rectifier

Bridge rectifier uses four diodes and does not require a center tapped transformer. It is common in practical power supplies.

- Two diodes conduct in each half cycle.
- Output polarity remains same for both half cycles.
- With filter capacitor, output becomes smoother DC.

M4.04 5V/9V regulated power supply

Use regulator IC such as 7805 or 7809. A rectifier and filter must provide sufficient input voltage to the regulator.

Regulated output: 7805 -> 5 V, 7809 -> 9 V

- ✓ Check regulator pinout before wiring.
- ✓ Use input and output capacitors if required.
- ✓ Measure output voltage without load and with load.
- ✓ Watch heating of regulator IC.

Regulated DC power supply block diagram.

M4.05 Logic gate truth table verification

Verify output for all input combinations using logic gate ICs or trainer kit.

A	B	AND	OR	NAND	NOR	XOR
0	0	0	0	1	1	0
0	1	0	1	1	0	1
1	0	0	1	1	0	1
1	1	1	1	0	0	0

Rectifier practical mistakes

- Connecting electrolytic capacitor in reverse polarity.
- Using diode with wrong orientation.
- Connecting CRO ground wrongly and shorting a point.
- Expecting perfect DC without filter.
- Forgetting to calculate ripple factor.

Record and observation templates

Rectifier observation table

Rectifier	Filter	Input Vrms	Output Vdc	Ripple voltage	Ripple factor	Waveform remarks
Half wave	No					
Half wave	Yes					
Center tapped	No					

Rectifier	Filter	Input Vrms	Output Vdc	Ripple voltage	Ripple factor	Waveform remarks
Bridge	Yes					

CRO observation table

Waveform	Volts/div	Vertical divisions	Vpp	Time/div	Horizontal divisions/cycle	T	f
Sine							
Square							

Formula bank and practical shortcut notes

Basic circuit

$V = I R$

$P = V I = I^2R = V^2/R$

Energy = P x t

AC quantities

$V_{rms} = V_m / \text{sqrt}(2)$

$I_{rms} = I_m / \text{sqrt}(2)$

Frequency $f = 1/T$

Rectifier

Ripple factor $r = V_{ac\ rms}/V_{dc}$

HWR ideal $r \approx 1.21$

Full wave/bridge ideal r ≈ 0.482

Meter selection rule

When expected value is unknown, start from the highest range and reduce gradually for better resolution. This protects the instrument.

Polarity rule

DC meters, electrolytic capacitors, diodes, LEDs and regulator ICs are polarity sensitive. Check before switching ON.

Viva question bank with direct answers

CO1 viva - meters and circuits

1. **Why is an ammeter connected in series?** Because it must carry the same circuit current.
2. **Why is a voltmeter connected in parallel?** Because voltage is measured across two points.
3. **What is multiplication factor?** Factor used to convert scale reading into actual reading for selected range.
4. **What happens if one lamp opens in series?** Entire series circuit opens.
5. **What is the role of fuse?** Protection against excess current.

CO2 viva - power and energy

1. **What is one unit of energy?** One kilowatt-hour.
2. **What are wattmeter coils?** Current coil and pressure coil.
3. **How is power calculated in DC circuit?** P = V x I.
4. **Why should battery terminals be clean?** To reduce contact resistance and voltage drop.
5. **Which water is used in lead acid battery?** Distilled water.

CO3 viva - components and CRO

1. **What is CRO used for?** To display and measure electrical waveforms.
2. **How to find frequency from CRO?** Measure T and use f = 1/T.
3. **What is diode testing?** Forward low resistance and reverse high resistance check.
4. **What is breadboard?** Solderless prototyping board.
5. **What is LED?** Light Emitting Diode.

CO4 viva - rectifiers and logic

1. **What is rectification?** Conversion of AC to DC.
2. **What is ripple?** AC component present in rectifier output.
3. **Why is filter capacitor used?** To reduce ripple and smooth the output.
4. **What is bridge rectifier?** Full wave rectifier using four diodes.
5. **What is universal gate?** NAND and NOR gates.

Lab test preparation: model practical paper

This course has Lab Test I and Lab Test II. For a practical exam, preparation must include wiring, observation, calculation and viva.

Lab Test I - sample tasks

1. Assemble a source-switch-fuse-load circuit and test operation.
2. Measure voltage and current in a DC circuit using correct range.
3. Verify voltage and current division in series and parallel lamp circuits.
4. Measure power using wattmeter.
5. Measure energy using energy meter.
6. List maintenance steps for lead acid battery.

Lab Test II - sample tasks

1. Identify passive components and write their values.
2. Measure amplitude, time period and frequency using CRO.
3. Test diode, LED, resistor and capacitor.
4. Construct half wave rectifier and calculate ripple factor.
5. Construct bridge rectifier with filter and plot waveform.
6. Verify truth table of basic logic gates.

Practical answer writing format

Result format example: The circuit was assembled and tested successfully. The measured voltage was ___ V and current was ___ A. The calculated power was ___ W. The result agrees with theoretical expectation within practical tolerance.

> Result □□□□□□□□□□□ simply “verified” □□□□□ □□□□□□ □□□□□□□□□. Measured value + unit + conclusion □□□□.

Text, reference and online resources

Type	Resource
T1	B. L. Theraja and A. K. Theraja, Textbook of Electrical Technology: Part 1 - Basic Electrical Engineering in S. I. Units, S. Chand Publication, 2012.
T2	N. N. Bhargava, D. C. Kulshreshtha and S. C. Gupta, Basic Electronics and Linear Circuits, Tata McGraw Hill Education, 1983.
R3	A. Anand Kumar, Fundamentals of Digital Circuits, Prentice Hall India Pvt. Ltd., 2014.
Online	Electrical4U; NPTEL/Digimat electrical and electronics lecture resources listed in syllabus.